Exercises 5 – Synchronous Motor

1. A 480-V 100-kW 0.85-PF leading 50-Hz six-pole Y-connected synchronous motor has a synchronous reactance of 1.5 Ω and a negligible armature resistance. The rotational losses are also to be ignored. This motor is to be operated over a continuous range of speeds from 300 to 1000 r/min, where the speed changes are to be accomplished by controlling the system frequency with a solid-state drive.
   a. Determine the supply frequency at the lowest speed and highest speed?
   b. How large is E_A at the motor's rated conditions?
   c. What is the maximum power the motor can produce at the rated conditions?

2. A 440-V three-phase Y-connected synchronous motor has a synchronous reactance of 1.5Ohm per phase. The field current has been adjusted so that the torque angle is 28° when the power supplied by the generator is 90 kW.
   a. What is the magnitude of the internal generated voltage $\mathbf{E}_A$ in this machine?
   b. What are the magnitude and angle of the armature current in the machine? What is the motor's power factor?
   c. If the field current remains constant, what is the absolute maximum power this motor could supply?

3. A 2300-V 400-hp 60-Hz eight-pole Y-connected synchronous motor has a rated power factor of 0.85 leading. At full load, the efficiency is 85 percent. The armature resistance is 0.4 Ohm, and the synchronous reactance is 4.4 Ohm. Find the following quantities for this machine when it is operating at full load:
   a. Output torque
   b. Input power
   c. Mechanical speed
   d. E_A
   e. P_{conv}
   f. $P_{mech} + P_{core} + P_{stray}$

4. A 480-V, 60 Hz, 400-hp 0.8-PF-leading eight-pole Δ-connected synchronous motor has a synchronous reactance of 1.0 Ω and negligible armature resistance. Ignore its friction, windage, and core losses for the purposes of this problem.

a. If this motor is initially supplying 400 hp at 0.8 PF lagging, what are the magnitudes and angles of $\mathbf{E}_A$ and $\mathbf{I}_A$?
b. How much torque is this motor producing? What is the torque angle δ?
c. If $|\mathbf{E}_A|$ is increased by 15 percent, what is the new magnitude of the armature current? What is the motor's new power factor?